

Chapter 15: Fun with Magnets

Textbook Exercise Solutions & Answers

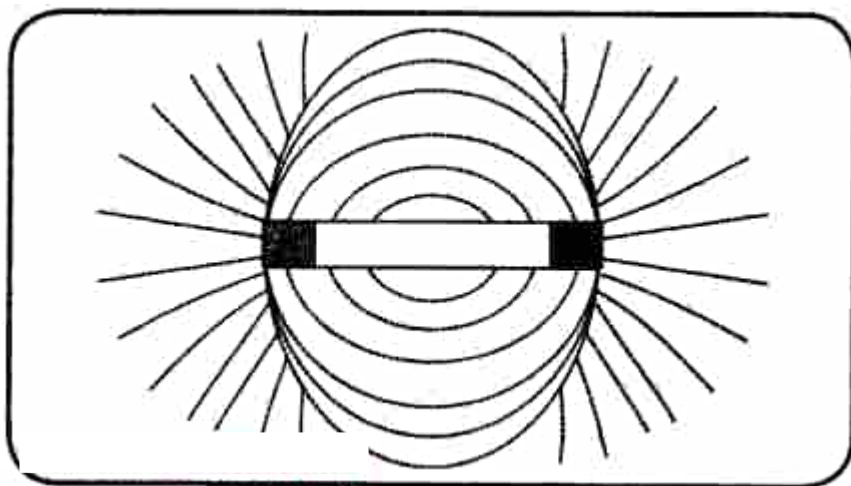
1. How will you do this?

(a) Determine whether a material is magnetic or non-magnetic.

1. To determine whether the material is magnetic or non-magnetic, a magnet is moved over it.
2. If the material sticks to the magnet, it is called magnetic material.
3. If the material does not stick to the magnet, it is non-magnetic.

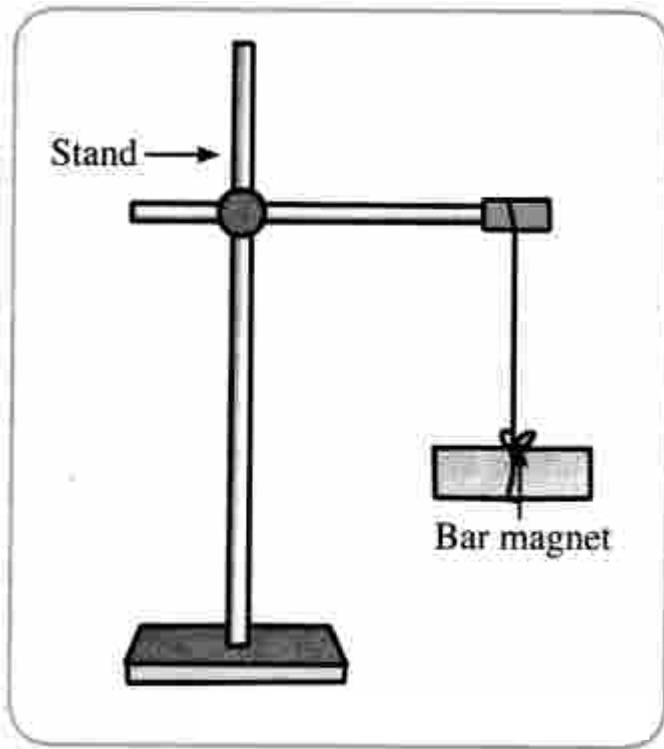
(b) Explain that a magnet has a certain magnetic field.

1. The space around a magnet in which the magnetic force is active is called the magnetic field.
2. Place a white paper on a drawing board and place a bar magnet in the middle of the paper.
3. Spread the iron filings on the sheet and gently tap the sheet.
4. The iron filings arrange around the magnet in definite curved lines forming a symmetric pattern.
5. The lines are closer to each other near the poles and less crowded in the middle region around the magnet.
6. Beyond a particular region, the iron filings, will not get attracted.
7. The region where iron filings are attracted is the magnetic field of the magnets.



(c) Find the north pole of a magnet.

1. Take a bar magnet. Tie a thread to the centre of a bar magnet and hang it from a stand.
2. Note the direction in which the magnet settles and turn it around again.
3. Allow it to settle and note the direction.
4. The end of the magnet that points to the north is called the north pole, while the end that points to the south is called the south pole.
5. The north pole is indicated by 'N' and the south pole by 'S'.



2. Which magnet will you use?

(a) Iron is to be separated from a trash.

1. Sharp and heavy iron scrap material is attached to a big disc.
2. The disc is a magnet and all scrap is attracted to it.
3. It is not possible to create, store such a big size magnet. Therefore magnetism is induced in the disc with the help of electricity.

An electromagnet is used which is attached to a crane for loading and unloading, transporting scrap and loose iron material from a trash.

(b) You are lost in a forest.

1. If we are lost in a forest, we should take help of a mariner's compass which will help us to find the directions while travelling through unknown regions.
2. If mariner's compass is not available, a bar magnet when suspended in the centre will rest in north-south direction.

(c) A window shutter opens and shuts continuously in the wind.

A bar magnet can be attached to the window pane so that the window will be closed tight during strong winds also.

3. Fill in the blanks with appropriate word.

(a) If a bar magnet is hung by a thread tied at its centre, its north pole becomes steady in the direction of the pole of the earth. (south, north, east, west)

North

(b) If a bar magnet is cut into equal pieces by cutting it at right angles to its axis at two pieces bar magnets are formed, and a total of poles are formed. (6,3,2)

3, 6

(c) There is a repulsion between the poles of a magnet and attraction between its poles. (opposite, like.)

like, opposite

(d) When magnetic material is taken close to a magnet, the material acquires (permanent magnetism, induced magnetism, temporary, magnet keeper)

induced magnetism

(e) If a magnet attracts a piece of metal, that piece must be made of(any other metal but iron, magnetic material or iron, non-magnetic material, electromagnets)

magnetic material or iron

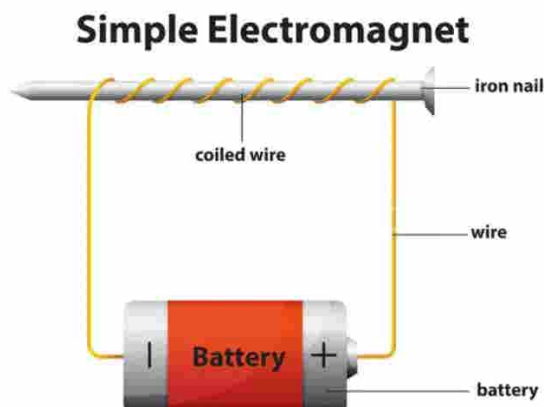
(f) A magnet remains steady in a direction. (east-west, north-south,)

north-south

4. Write the answers in your words.

(a) How is an electromagnet made?

1. To make an electromagnet we need the following apparatus; An iron nail of 10 cm length, 1 metre long insulated copper wire, a battery cell, pins.
2. Wind the copper wire around the nail as shown in the figure. Connect the free ends of the wire to the two terminals of a cell through a plug key.
3. Close the key to complete the circuit.
4. Bring small pins near the tip of the nail and observe.
5. When the circuit is completed, the iron pins are attracted by the nail and hence, they stick to the nail.



6. When the circuit is broken, the pins fall off.
7. A magnet is prepared by passing an electric current through an insulated wire wound around the iron nail. This is an electromagnet.
8. When the current is allowed to pass, the nail becomes a magnet and attracts pin / pins stick to it.
9. When the current is put off the nail does not behave as a magnet and therefore, pins fall off.
10. The magnetism is temporary in the case of an electromagnet.

(b) Write the properties of a magnet.

Magnet possess following properties/ characteristics.

1. Magnet always settles in the north-south direction.
2. The magnetic force is concentrated at the two ends or poles of a magnet.
3. If a magnet is divided into two parts, two independent magnets are formed. It means that the two poles of a magnet cannot be separated from each other.
4. A magnetic material acquires magnetism when placed near a magnet. This magnetism is called induced magnetism.
5. There is repulsion between like poles of a magnet, while there is attraction between the opposite poles.

(c) What are the practical uses of a magnet?

1. Magnets are materials to which objects made from iron, nickel, cobalt are attracted. But man explored magnets and its properties and made his life comfortable.
2. Permanent Magnets: are used in caps of pin holders, doors of fridges, doors of cupboards etc.
3. Temporary magnets: Electromagnets are used in electric bells, circuit of various machines, ATM card swipe machines, MRI- Magnetic Resonance Image, loudspeakers, electric cranes, microphones, Mariner's compasses, etc.